Software Design Analysis Report

A. Project Details

Project title: AI-Based Interview Preparation Platform

Submitted By:

1. Himanshu Rawat – 202401100100120
2. Himanshu Chaudhry – 202401100100118
3. Devesh Mishra – 202401100100096
4. Dharmendra Patel – 202401100100097
5. Krishnkant Gupta – 202401100100137

Section: CS-B

Project objective (2–3 lines):

The objective of the AI-Based Interview Preparation Platform is to help students and job seekers prepare for technical, coding, and HR interviews. The platform provides interview questions, AI-assisted answer evaluation, personalized feedback, and performance reports to improve interview preparation.

Technology stack:

- Frontend: React.js
- Backend: Python FastAPI
- Database: MongoDB
- AI Integration: External AI API
- Programming Languages: Python, JavaScript

Note: Ye stack aapke actual project implementation ke according verify kar lena.

B. Module Identification and Modularization

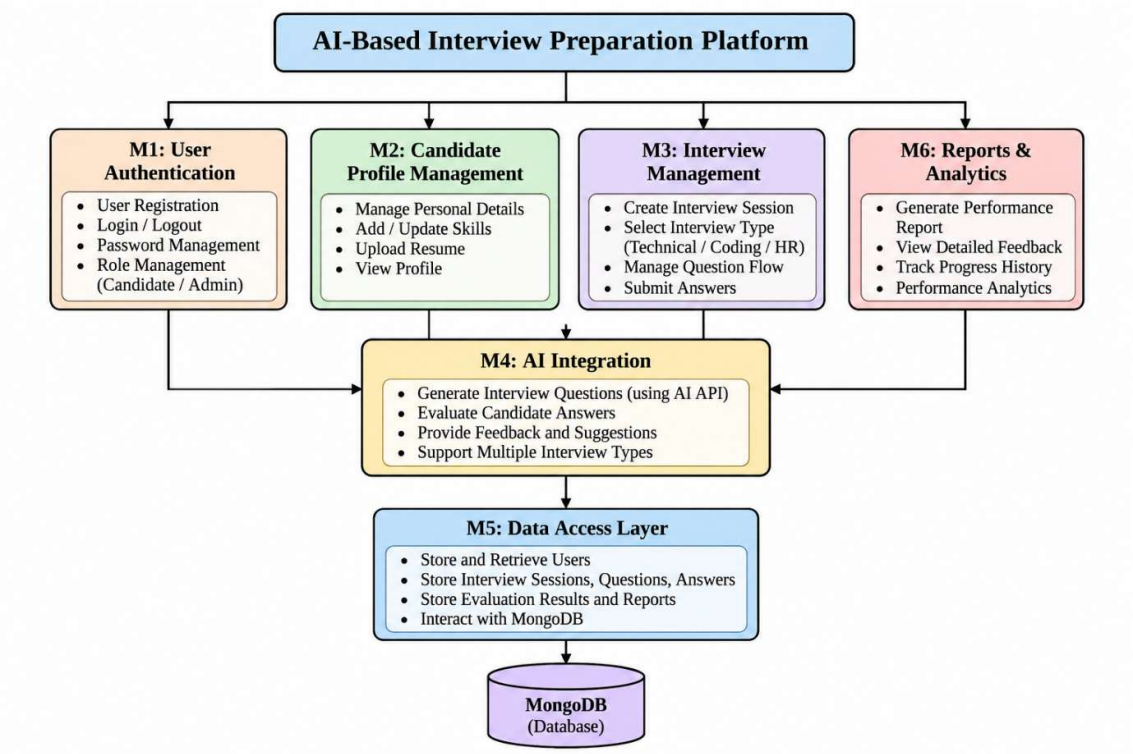
Total number of modules: 6

The project is divided into six modules to separate responsibilities, simplify development, improve testing, and make the system easier to maintain.

Module ID	Module name	Main responsibility	Inputs / Outputs	Depends on
M1	User Authentication	User registration, login and authentication	Credentials → authentication result	M5
M2	Candidate Profile	Manage skills, experience and resume details	Profile data → updated profile	M5

M3	Interview Management	Create sessions and manage interview questions and answers	Interview type and answers → session data	M1, M4, M5
M4	AI Integration	Generate questions and evaluate answers using an AI service	Prompt/answer → question, score and feedback	External AI API
M5	Data Access	Store and retrieve users, sessions, answers and results	Data → database records	MongoDB
M6	Reports and Analytics	Generate performance summaries and display results	Evaluation records → performance report	M3, M5

Module hierarchy / decomposition diagram



Explanation:

The system is divided into modules according to their responsibilities. Authentication manages user access, the profile module manages candidate information, and interview management controls the interview workflow. AI integration generates questions and

evaluates answers. The data access module handles persistence, while reports and analytics summarize interview performance.

C. Design Strategy

Selected strategy: ☑ Top-Down ☑ Hybrid ☐ Bottom-Up

Justification (4–6 lines):

The project follows a top-down design strategy by starting with the complete interview preparation system and dividing it into smaller functional modules. Each module is assigned a specific responsibility. This makes the architecture easier to understand and manage. A hybrid strategy is also suitable because reusable components, such as database access and AI integration, can be developed and tested independently. Finally, all modules are integrated to form the complete application.

Evidence:

The system was decomposed into authentication, candidate profile, interview management, AI integration, data access, and reporting modules. Each module has defined inputs, outputs, and dependencies. The modules can be developed independently and integrated through clearly defined interfaces.

Complete System

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Module Identification

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Functional Decomposition

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Interface Definition

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Individual Implementation

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Module Integration

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Complete Platform

D. Design Approach

Selected approach: ☐ Function-Oriented ☑ Object-Oriented ☑ Mixed

Justification (4–6 lines):

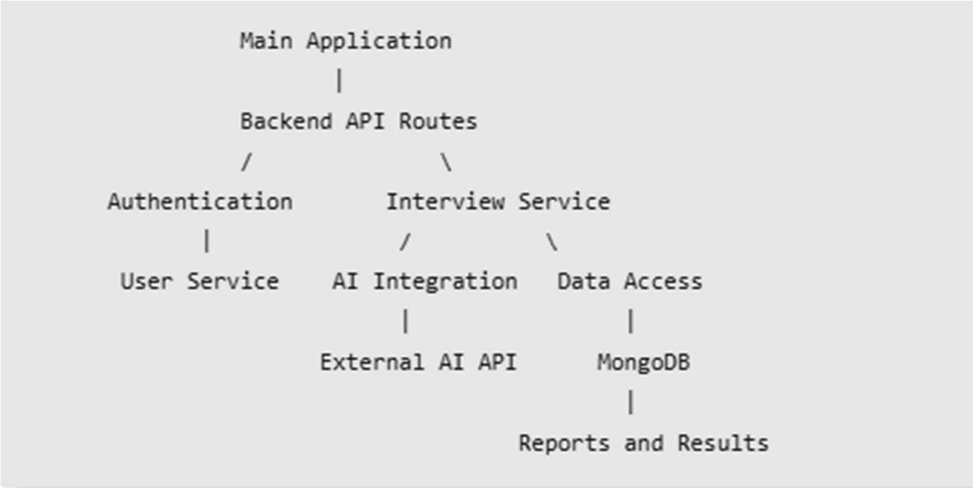
The project follows an object-oriented approach because it contains entities such as User, Candidate, InterviewSession, Question, Answer, Evaluation, and PerformanceReport. These

entities represent the main components of the system. The application also uses functions and API endpoints to perform operations such as login, question generation, answer submission, and report retrieval. Therefore, a mixed approach combining object-oriented modelling with function-based operations is suitable for this project.

Main classes / objects:

- User
- Candidate
- InterviewSession
- Question
- Answer
- Evaluation
- AIService
- InterviewRepository
- PerformanceReport

Structure chart



Explanation:

The API routes receive requests from the frontend and forward them to the appropriate service. Authentication handles login and registration, while interview management coordinates questions and answers. The AI integration service communicates with the AI provider. The data access layer manages database operations, and the report service generates performance summaries.

E. Coupling Analysis

Coupling refers to the degree of dependency between software modules. Low coupling is desirable because changes in one module should have minimal impact on other modules.

Module pair	Interaction / shared data	Coupling type / level	How to reduce dependency
Interview Management and AI Integration	Questions, answers, evaluation results	Data coupling; low to moderate	Use a common AI service interface and standardized response format.
Interview Management and Data Access	Session IDs, questions, answers and session status	Data coupling; low	Use repository methods instead of directly accessing database internals.
Authentication and Data Access	User ID, email and password hash	Data coupling; low	Expose focused methods for user lookup and authentication-related data.
Reports and Data Access	Evaluation records and score summaries	Data coupling; low	Retrieve only the data required for report generation.

Conclusion:

The proposed design aims to achieve low coupling by exchanging structured data through well-defined interfaces. This reduces dependencies between modules and makes the system easier to modify, test, and maintain.

F. Cohesion Analysis

Cohesion refers to how closely the responsibilities within a module are related. High cohesion is desirable because each module should focus on one well-defined task.

Module	Main responsibility	Cohesion type / level	Improvement, if needed
Authentication Service	Register users, validate credentials and manage authentication	Functional cohesion; high	Keep profile management and interview logic separate.
Interview Management	Create sessions and coordinate question-answer progression	Sequential/functional cohesion; high	Keep AI-provider-specific logic outside this module.
AI Integration	Send prompts and answers to the AI provider and normalize responses	Functional cohesion; high	Separate provider adapters if multiple providers are supported.
Data Access	Perform database read and write operations	Functional cohesion; high	Organize database operations by entity or repository.

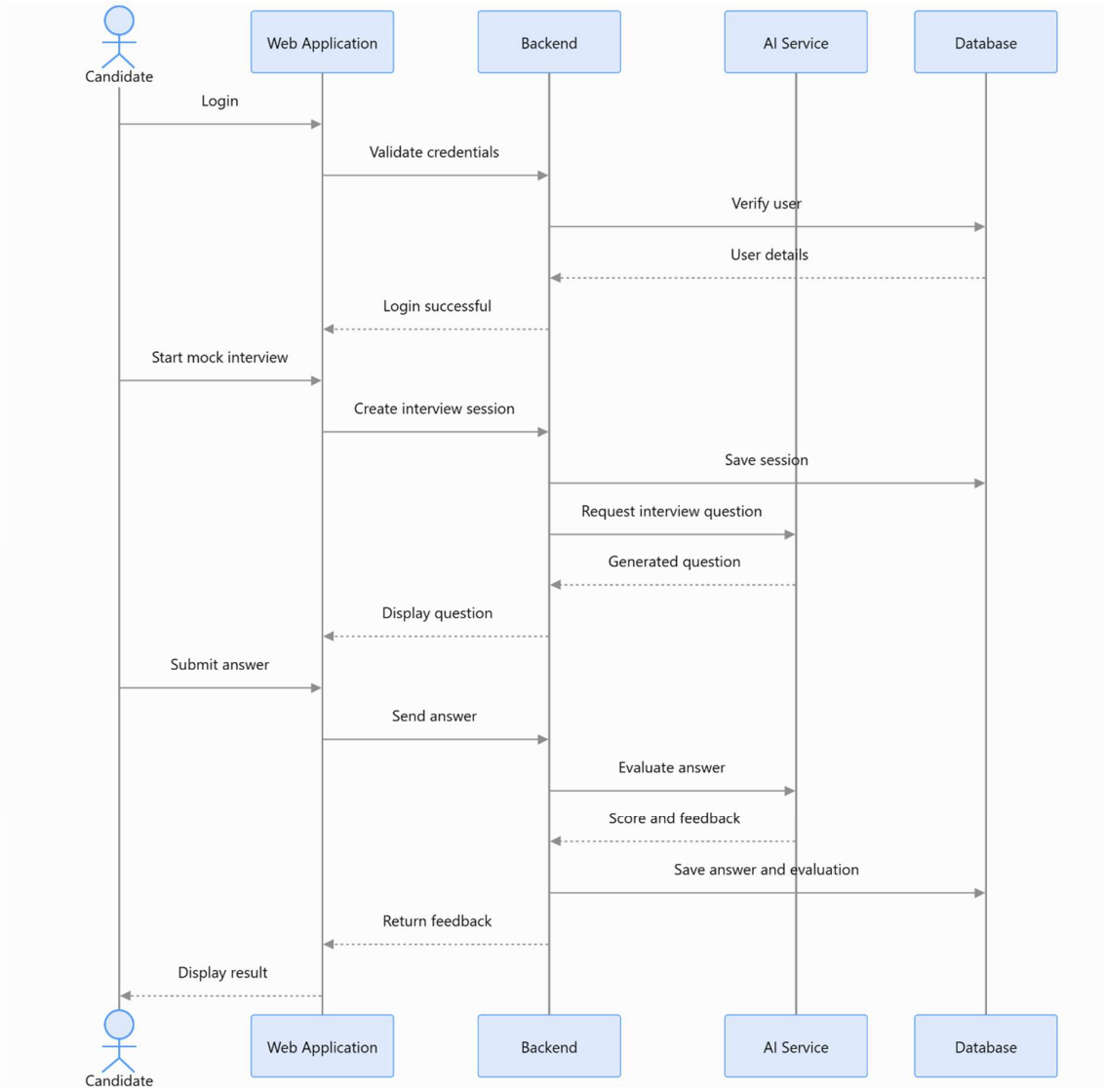
Reports and Analytics	Aggregate scores and prepare performance summaries	Functional cohesion; high	Keep report generation separate from interview session management.

Conclusion:

The proposed modular design aims to achieve high cohesion by assigning each module a clear responsibility. This improves readability, testing, maintainability, and reusability.

G. Module Interaction and Design Quality

Sequence diagram



Explanation:

1. The candidate logs into the platform through the frontend.
2. The backend verifies the user's credentials using the database.
3. The candidate starts a mock interview, and the backend creates a session.

4. The AI service generates or retrieves an interview question.
5. The candidate submits an answer through the frontend.
6. The backend sends the answer to the AI service for evaluation.
7. The evaluation result is stored in the database.
8. The candidate receives feedback and can view the performance report.

Design quality observations

1. Maintainability: Separating authentication, interview management, AI integration, and reporting makes changes easier to implement.

2. Testability: Each module can be tested independently. Mock AI responses can be used to test the interview workflow without making real API calls.

3. Reliability: The application should validate user inputs, handle AI API failures, and manage database errors appropriately.

4. Security: Passwords should be stored as hashes, API keys should be protected using environment variables, and access to interview history and reports should be authorized.

5. Limitations: AI-generated feedback may be inaccurate or inconsistent. It should be treated as guidance rather than a guaranteed assessment of interview performance.

Conclusion

The AI-Based Interview Preparation Platform uses modular design to separate authentication, candidate profile management, interview management, AI integration, database access, and reporting. The proposed architecture aims to maintain low coupling and high cohesion. Applying these software design principles improves maintainability, testability, reliability, and extensibility.

Result: The software design analysis was completed by identifying the major modules, selecting a design strategy and approach, and analyzing coupling, cohesion, and module interactions.